

# The one-stop operating handbook for ShadowTraceAI.

Why the problem matters, what the copilot delivers, how the Snowflake system works, and exactly how to build, operate, control, test, demonstrate, and evolve it—from synthetic prototype to production-ready design.

Domain-Specific AI Copilot

CASE-C003 · 92/100 CRITICAL

Windows PowerShell

Synthetic data only

|                                                             |                                                              |
|-------------------------------------------------------------|--------------------------------------------------------------|
| <div>DOCUMENT</div> <div>Integrated Solution Handbook</div> | <div>VERSION</div> <div>2.0</div>                            |
| <div>STATUS</div> <div>Prototype + target design</div>      | <div>OWNER</div> <div>ShadowTraceAI team</div>               |
| <div>AUDIENCE</div> <div>Business, risk, engineering</div>  | <div>CLASSIFICATION</div> <div>Synthetic demonstration</div> |

01 · ORIENTATION

## Executive overview

ShadowTraceAI is a Snowflake-powered AML intelligence copilot focused on modern-slavery and human-trafficking indicators. It connects transactions, KYC, devices, external intelligence, and document metadata; detects network-aware typologies in SQL; and converts evidence into an explainable, human-controlled case decision.

### 5

#### Typologies

Wage harvesting, account control, spending anomaly, geographic risk, and sector risk.

92

## Demo score

CASE-C003 reaches CRITICAL through visible additive logic, not a case-ID override.

8

## App tabs

From overview and network evidence through reviewer decision and audit trail.

Seed CSVs



Snowflake tables



Typology views



Risk + brief



Streamlit



Human decision + audit

Important operating model

The application can run in your local browser, but Snowflake remains the database. A fully offline database mode is intentionally not used.

## Document hierarchy

### READER QUESTION

### HANDBOOK LAYER

### DECISION ENABLED

Why should we invest?

Business proposition and measurable outcomes

Sponsor / funding decision

What will the solution do?

Scope, personas, requirements, controls

Product and risk approval

How is it designed?

SDD and DSD

Architecture and engineering approval

| READER QUESTION     | HANDBOOK LAYER                       | DECISION ENABLED          |
|---------------------|--------------------------------------|---------------------------|
| How is it operated? | SOP, RACI, deployment runbook        | Service readiness         |
| How do we prove it? | Tests, demo, KPIs, evidence register | Acceptance and submission |

02 • WHY

# Modern-slavery risk is a network problem hidden inside fragmented evidence.

Individual transactions can look ordinary. Suspicion becomes visible only when employer payroll, rapid onward transfers, shared control signals, weak KYC, risky sectors, recruitment corridors, documents, and external intelligence are evaluated together.

50M

### People affected

The ILO estimated that 50 million people were living in modern slavery in 2021, including 28 million in forced labour.

\$236B

### Illegal annual profit

The ILO estimated annual illegal profits from forced labour in the private economy at US\$236 billion.

1 ≠ proof

### Signals need context

FATF guidance stresses that one indicator alone is unlikely to confirm trafficking-related money laundering; combinations and context matter.

# Current-state failure pattern

|                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data                                                                                                                                                            |
| Transactions, KYC, devices, documents, watchlists, and media are reviewed in separate tools or queues.                                                          |
| Detection                                                                                                                                                       |
| Generic AML rules may not encode wage harvesting, shared account control, low personal spending, worker networks, and labour-intensive sector context together. |
| Investigation                                                                                                                                                   |
| Analysts manually reconstruct relationships, repeat searches, and write narrative explanations with inconsistent evidence coverage.                             |
| Decision                                                                                                                                                        |
| Risk reasoning, reviewer rationale, evidence gaps, and final outcomes can be difficult to trace as one chronology.                                              |
| Management                                                                                                                                                      |
| Without standard measures, leaders cannot reliably compare handling time, evidence completeness, false positives, or control adherence.                         |

- 1. ILO — 50 million people worldwide in modern slavery
- 2. ILO — Annual forced-labour profits estimated at US\$236 billion
- 3. FATF — Money-laundering risks from trafficking and migrant smuggling
- 4. FATF/APG — Financial flows from human trafficking

## 03 · WHAT

One governed evidence plane turns scattered alerts into defensible decisions.

The proposition is not “AI replaces the investigator.” It is “Snowflake assembles the evidence, SQL explains the risk, and a reviewer makes a better controlled decision.”

**Connect**

Unify transaction, customer, device, intelligence, and document evidence around a case and connected-account graph.

**Explain**

Calculate a reproducible 0–100 score with visible typology contributions, evidence strength, top factors, and route.

**Act safely**

Present an investigation brief and recommended decision while preserving named human review and a complete audit trail.

**Value pillars**

| PILLAR                 | OPERATIONAL CHANGE                                                                        | HOW VALUE IS MEASURED                            |
|------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------|
| Analyst productivity   | Less manual evidence collection and narrative assembly                                    | Active handling minutes per case                 |
| Evidence consistency   | Five typologies and required sources assessed systematically                              | Evidence-completeness rate                       |
| Network awareness      | Employer, worker, controller, funnel, device, and funds-flow relationships shown together | Connected entities identified per true case      |
| Decision defensibility | Score, factors, missing evidence, rationale, and chronology retained                      | Decisions with complete audit lineage            |
| Safer AI adoption      | Deterministic scoring works without generative AI; narratives are grounded and reviewable | Explainability coverage and human-gate adherence |

Domain coverage: individual and corporate signals

| TYPOLOGY         | INDIVIDUAL-ACCOUNT MANIFESTATION                                           | CORPORATE-ACCOUNT MANIFESTATION                                                              |
|------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Wage harvesting  | Employer credit followed by rapid bulk transfer to a third party           | Payroll recipients exceed declared employee count or registry profile                        |
| Account control  | Shared device, IP, contact, address, or handler across unrelated holders   | Directors/owners differ from observed operators or controller contacts                       |
| Spending anomaly | Little food, rent, transport, or ordinary personal spend despite income    | Cash or payment behaviour conflicts with stated business model                               |
| Geographic risk  | Recruitment corridors and rapid foreign cash-out inconsistent with profile | Payments to exploitation-linked entities or locations without business rationale             |
| Sector risk      | Income tied to labour-intensive sectors with other control indicators      | Construction, agriculture, hospitality, labour supply, or similar activity plus KYC mismatch |

Outcome statement

Reduce time-to-insight and variation in AML investigations while improving evidence quality, reviewer accountability, and audit reconstruction.

04 · WHO

Every stakeholder receives a different, traceable outcome.

| PERSONA                         | PRIMARY NEED                          | SHADOWTRACEAI OUTCOME                                                | DECISION RIGHT                                         |
|---------------------------------|---------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------|
| AML investigator                | Understand the case quickly           | Connected evidence, signals, network, score, brief, missing evidence | Recommend or record case disposition according to role |
| Senior reviewer / MLRO delegate | Make a defensible escalation decision | Full rationale, source evidence, confidence, audit chronology        | Escalate, request evidence, monitor, or close          |

| PERSONA                       | PRIMARY NEED                            | SHADOWTRACEAI OUTCOME                                              | DECISION RIGHT                         |
|-------------------------------|-----------------------------------------|--------------------------------------------------------------------|----------------------------------------|
| Financial-crime product owner | Operate a consistent service            | Standard workflow, KPIs, exceptions, backlog, and control evidence | Prioritise roadmap and acceptance      |
| Data / platform engineer      | Deliver reliable governed data products | Typed model, ordered SQL build, views, tests, deployment runbook   | Technical release and recovery         |
| Model risk / compliance       | Validate logic and human oversight      | Visible weights, thresholds, prompt boundaries, test evidence      | Approve scoring/prompt changes         |
| Audit / assurance             | Reconstruct who knew what and when      | Linked decisions, actors, objects, timestamps, and metadata        | Challenge control design and operation |

05 · BOUNDARY

**Prototype scope is focused; production design is explicit.**

**In scope now** ● Implemented

- Six synthetic seed datasets and two cases
- Ten requested operational tables plus brief persistence
- Five typology views and a case-network view
- Explainable score, evidence strength, route, and brief
- Eight-tab Streamlit reviewer experience
- Four reviewer decisions with transactional audit write
- SQL acceptance tests and CoCo evidence workflow

Production target ● Design

- Controlled source ingestion, CDC, and data contracts
- Enterprise identity, RBAC, masking, retention, and legal hold
- Case assignment, queues, SLAs, comments, and attachments
- Threshold calibration and outcome-based monitoring
- Prompt/model governance and evaluation
- Operational observability, incident response, and DR
- Jurisdiction-specific SAR preparation interfaces

Explicitly out of scope

- Autonomous SAR filing, customer restriction, account closure, or law-enforcement reporting.
- Determining that trafficking or exploitation has occurred.
- Use of real customer, victim, employee, or confidential bank data in the hackathon prototype.
- Production claims about time or cost reduction before a measured pilot.

Core assumptions

All identities and evidence are synthetic; rules illustrate a defensible framework rather than universal legal thresholds; business benefits are hypotheses to measure through a controlled pilot.

06 · REQUIREMENTS BASELINE

Business, functional, control, and service requirements are traceable.

| ID    | REQUIREMENT                                                                 | ACCEPTANCE EVIDENCE                             | STATUS      |
|-------|-----------------------------------------------------------------------------|-------------------------------------------------|-------------|
| BR-01 | Provide one case view across structured and document evidence               | Evidence and brief tabs return linked case data | Implemented |
| BR-02 | Explain modern-slavery typology risk at case and network level              | Five signal views and network graph             | Implemented |
| FR-01 | Load accounts, transactions, KYC, device/IP, external, and document records | Seed COPY commands and row validation           | Implemented |

| ID     | REQUIREMENT                                                              | ACCEPTANCE EVIDENCE                          | STATUS      |
|--------|--------------------------------------------------------------------------|----------------------------------------------|-------------|
| FR-02  | Calculate 0–100 score, band, evidence strength, factors, and route       | vw_case_risk_summary                         | Implemented |
| FR-03  | Generate a grounded investigation brief with missing evidence            | vw_case_intelligence_brief                   | Implemented |
| FR-04  | Allow four controlled reviewer decisions with rationale                  | Reviewer form and decision history           | Implemented |
| CR-01  | Never file a SAR autonomously                                            | Human acknowledgement and procedure boundary | Implemented |
| CR-02  | Write reviewer decision and audit event atomically                       | Stored-procedure transaction and linked IDs  | Implemented |
| NFR-01 | Preserve reproducibility without generative AI                           | Deterministic SQL score and brief            | Implemented |
| NFR-02 | Enforce least privilege, masking, retention, and environment segregation | Production security design and control tests | Target      |
| NFR-03 | Expose pipeline freshness, failures, data quality, and usage telemetry   | Operational dashboards and alerting          | Target      |

### Requirement-to-proof traceability

| USER NEED               | SNOWFLAKE PROOF OBJECT             | APPLICATION PROOF | TEST / CONTROL               |
|-------------------------|------------------------------------|-------------------|------------------------------|
| See wage diversion      | vw_wage_harvesting_signals         | Typology Signals  | Five-typology assertion      |
| See coordinated control | device_ip_signals, vw_case_network | Account Network   | Seven shared-access accounts |

| USER NEED           | SNOWFLAKE PROOF OBJECT  | APPLICATION PROOF | TEST / CONTROL                          |
|---------------------|-------------------------|-------------------|-----------------------------------------|
| Understand 92       | vw_case_risk_summary    | Risk Score        | Exact score/band/evidence assertions    |
| Make a decision     | sp_record_case_decision | Reviewer Decision | Allowed-value and atomic-write controls |
| Reconstruct history | audit_events            | Audit Trail       | Decision ID linkage query               |

07 · BEFORE YOU BEGIN

# Prerequisites

## Local workstation

- ☐ Python 3.10 or newer
- ☐ PowerShell and Git
- ☐ Project cloned to the local machine
- ☐ Network access to your Snowflake account

## Snowflake

- ☐ Active Snowflake account
- ☐ Role that can create database, schema, stage, tables, views, procedure, and Streamlit
- ☐ Running warehouse such as COMPUTE\_WH
- ☐ Optional Cortex role/model access

```
python --version
git --version
Get-ExecutionPolicy
```

# Prepare the Snowflake account

Use an existing warehouse or create one in Snowsight. For a hackathon sandbox, start with a small warehouse and suspend it when not in use.

## 1 Confirm context in Snowsight

```
SELECT CURRENT_ACCOUNT(), CURRENT_USER(), CURRENT_ROLE(),  
CURRENT_WAREHOUSE();  
SHOW WAREHOUSES;
```

## 2 Optionally create a dedicated warehouse

```
CREATE WAREHOUSE IF NOT EXISTS SHADOWTRACE_WH  
  WAREHOUSE_SIZE = 'XSMALL'  
  AUTO_SUSPEND = 60  
  AUTO_RESUME = TRUE  
  INITIALLY_SUSPENDED = TRUE;
```

If you use this warehouse, change `COMPUTE_WH` to `SHADOWTRACE_WH` in the connection and deployment configuration.

### Cost control

Use auto-suspend and avoid leaving a warehouse running after the demo. Cortex calls may incur separate consumption.

# Create the local environment

## 1 Open the repository

```
cd C:\Users\manoj\Development\Github\ShadowTraceAI-Snowflake-CoCo
```

## 2 Create and activate a virtual environment

```
python -m venv .venv  
.\.venv\Scripts\Activate.ps1  
python -m pip install --upgrade pip  
pip install -r requirements.txt
```

If script execution is blocked, use `Set-ExecutionPolicy -Scope Process Bypass` for the current PowerShell process.

## 3 Install Snowflake CLI

```
pipx install snowflake-cli  
snow --version
```

If pipx is unavailable, install the CLI in the virtual environment with `pip install snowflake-cli`.

## 10 • AUTHENTICATION

# Configure and test the connection

**1**

## Add the Snowflake CLI connection

```
snow connection add  
snow connection test -c shadowtrace
```

Use shadowtrace as the connection name, or replace it consistently in every command below.

**2**

## Create local Streamlit secrets

Create `.streamlit/secrets.toml`:

```
[connections.snowflake]  
account = "YOUR_ACCOUNT_IDENTIFIER"  
user = "YOUR_USERNAME"  
password = "YOUR_PASSWORD_OR_TOKEN"  
role = "ACCOUNTADMIN"  
warehouse = "COMPUTE_WH"  
database = "SHADOWTRACE_AI"  
schema = "AML"
```

### Never commit credentials

`.streamlit/secrets.toml` is ignored by the repository. Prefer browser SSO or a scoped programmatic access token where your organization supports it.

## Connection smoke test

```
snow sql -c shadowtrace -q "SELECT CURRENT_USER(), CURRENT_ROLE(),  
CURRENT_WAREHOUSE(), CURRENT_VERSION()"
```

# Create and populate the application objects

Run the files in order. The seed loader uses relative file://data/seed/ ... paths, so remain in the repository root.

| ORDER | FILE                   | ACTIVITY                                     | EXPECTED RESULT                   |
|-------|------------------------|----------------------------------------------|-----------------------------------|
| 1     | 01_schema.sql          | Database, schema, tables, file format, stage | Core objects created              |
| 2     | 02_load_seed_data.sql  | Upload CSV files and load synthetic cases    | CASE-C003 and CASE-C010 available |
| 3     | 03_typology_views.sql  | Five typologies and network view             | Five C003 signals detected        |
| 4     | 04_risk_scoring.sql    | Risk summary and persisted score             | 92 / CRITICAL / STRONG            |
| 5     | 05_case_brief.sql      | Prompt, deterministic brief, timeline        | Investigation brief available     |
| 6     | 06_decisions_audit.sql | Transactional reviewer procedure             | Procedure created                 |

```
snow sql -c shadowtrace -f sql/01_schema.sql
snow sql -c shadowtrace -f sql/02_load_seed_data.sql
snow sql -c shadowtrace -f sql/03_typology_views.sql
snow sql -c shadowtrace -f sql/04_risk_scoring.sql
snow sql -c shadowtrace -f sql/05_case_brief.sql
snow sql -c shadowtrace -f sql/06_decisions_audit.sql
```

Optional Cortex activity

After the deterministic application works, execute sql/05b\_cortex\_brief\_optional.sql. Skip it if your region, role, or account lacks model access.

12 · QUALITY GATE

# Validate the data and risk logic

## 1 Run the automated SQL tests

```
snow sql -c shadowtrace -f tests/test_risk_logic.sql
```

Six rows should return **PASS**. The script raises an exception if any assertion fails.

## 2 Inspect the five detected typologies

```
SELECT signal_type, risk_points, explanation
FROM SHADOWTRACE_AI.AML.vw_all_typology_signals
WHERE case_id = 'CASE-C003' AND signal_detected
ORDER BY risk_points DESC;
```

## 3 Reconcile the score

```
SELECT case_id, risk_score, risk_band, evidence_strength,
       recommended_route, score_breakdown, top_risk_factors
FROM SHADOWTRACE_AI.AML.vw_case_risk_summary
ORDER BY risk_score DESC;
```

| FACTOR           | POINTS | EVIDENCE                                      |
|------------------|--------|-----------------------------------------------|
| Wage harvesting  | 25     | Five workers rapidly transfer most wages      |
| Account control  | 25     | Seven accounts share device and IP            |
| Spending anomaly | 14     | Minimal everyday spend and rapid depletion    |
| Geographic risk  | 12     | Cross-border recruitment and foreign cash-out |
| Sector risk      | 8      | Construction and incomplete ownership         |

| FACTOR                | POINTS | EVIDENCE                                              |
|-----------------------|--------|-------------------------------------------------------|
| External intelligence | 4      | Adverse-media/watchlist matches                       |
| Document evidence     | 4      | Payroll, contract, identity, accommodation indicators |
| Total                 | 92     | CRITICAL · STRONG                                     |

13 · LOCAL EXPERIENCE

# Run Streamlit locally

```
.\.venv\Scripts\Activate.ps1
streamlit run app/streamlit_app.py
```

Open <http://localhost:8501>. Select **CASE-C003**.

## Expected visual state

- Risk score: **92/100**
- Band: **CRITICAL**
- Evidence: **STRONG**
- Route: **Escalate to SAR**
- Five typology signals

## Eight tabs

Overview · Typology Signals · Account Network · Evidence · Risk Score · Case Intelligence Brief · Reviewer Decision · Audit Trail

### Local success condition

The browser renders CASE-C003 and switching between all eight tabs produces Snowflake-backed results without a connection error.

# Solution Design Document: Snowflake is the governed evidence and decision plane.

The design keeps data, detection, scoring, case intelligence, reviewer decisions, and audit lineage close together. The Streamlit layer presents and writes through governed Snowflake objects rather than implementing parallel business logic.

## Logical architecture



## Component responsibilities

| LAYER                 | RESPONSIBILITY                              | KEY OBJECTS                                      | FAILURE BOUNDARY                           |
|-----------------------|---------------------------------------------|--------------------------------------------------|--------------------------------------------|
| Ingestion             | Stage and type source records               | SEED_STAGE, CSV_SEED_FORMAT, core tables         | Reject malformed or contract-breaking data |
| Detection             | Detect evidence-backed typologies           | Five signal views                                | No case disposition; signals only          |
| Network               | Expose account and funds-flow relationships | vw_case_network                                  | No inferred guilt or victim determination  |
| Decision intelligence | Score, rank factors, recommend route        | vw_case_risk_summary                             | Recommendation is non-binding              |
| Narrative             | Assemble deterministic brief                | vw_case_intelligence_brief, optional AI_COMPLETE | Only supplied evidence; reviewer validates |

| LAYER      | RESPONSIBILITY                              | KEY OBJECTS                               | FAILURE BOUNDARY              |
|------------|---------------------------------------------|-------------------------------------------|-------------------------------|
|            | and optional AI prompt                      |                                           |                               |
| Experience | Present evidence and capture human decision | streamlit_app.py                          | No regulatory filing endpoint |
| Control    | Persist decision and audit chronology       | sp_record_case_decision, decisions, audit | Rollback on any write failure |

### Environment topology

| ENVIRONMENT | DATA                        | PURPOSE                                                  | PROMOTION GATE                                     |
|-------------|-----------------------------|----------------------------------------------------------|----------------------------------------------------|
| LOCAL       | Synthetic only              | Streamlit development against Snowflake DEV              | Compile and developer smoke test                   |
| DEV         | Synthetic / masked          | SQL integration and feature development                  | Automated tests                                    |
| TEST / UAT  | Representative masked cases | Calibration, control validation, investigator acceptance | Risk, product, security approval                   |
| PROD        | Controlled customer data    | Operational investigation service                        | Change record, rollback, monitoring, duty approval |

### Production non-functional design

#### Security and privacy

Federated identity, least privilege, row/column policies, masking, secure views, encryption, secrets management, purpose limitation, retention, and access review.

## Reliability

Idempotent ingestion, retry boundaries, freshness checks, release rollback, backup/retention controls, and documented recovery objectives.

## Performance

Case selection and core views should meet an agreed interactive-response SLO at expected concurrency and evidence volume.

## Observability

Pipeline failures, data quality, query latency, app errors, score distribution drift, decision mix, and human-gate exceptions become monitored service signals.

15 • DSD

# Detailed System Design: objects, grain, dependencies, and algorithms are explicit.

## Core data model

| OBJECT       | GRAIN / KEY                                 | PURPOSE                             | SUGGESTED PRODUCTION CLASSIFICATION |
|--------------|---------------------------------------------|-------------------------------------|-------------------------------------|
| accounts     | One row per account<br>· account_id         | Customer/account and sector context | Restricted customer data            |
| transactions | One row per transaction<br>· transaction_id | Funds-flow evidence                 | Restricted financial data           |

| OBJECT             | GRAIN / KEY                              | PURPOSE                                             | SUGGESTED PRODUCTION CLASSIFICATION |
|--------------------|------------------------------------------|-----------------------------------------------------|-------------------------------------|
| kyc_profiles       | One profile snapshot<br>· kyc_profile_id | Identity, expected activity, ownership              | Restricted personal/KYC data        |
| device_ip_signals  | One observation ·<br>signal_id           | Shared-control and access indicators                | Restricted telemetry                |
| external_watchlist | One source match ·<br>watchlist_id       | Sanctions-adjacent, registry, adverse-media context | Confidential intelligence           |
| document_evidence  | One document ·<br>document_id            | Metadata, extraction confidence, indicators         | Restricted evidence                 |
| case_alerts        | One active alert/case<br>· case_id       | Case header and assignment                          | Regulatory confidential             |
| case_risk_scores   | One score version ·<br>risk_score_id     | Persisted explanation and route                     | Regulatory confidential             |
| case_decisions     | One reviewer decision ·<br>decision_id   | Named disposition and rationale                     | Regulatory confidential             |
| audit_events       | One event ·<br>audit_event_id            | Actor/object/time/event lineage                     | Controlled audit record             |

Build dependency order



Explainable scoring algorithm

| SIGNAL                | DETECTION CONDITION IN PROTOTYPE                                   | POINTS | CALIBRATION OWNER               |
|-----------------------|--------------------------------------------------------------------|--------|---------------------------------|
| Wage harvesting       | At least three workers transfer $\geq 70\%$ onward within 24 hours | 25     | Financial-crime typology owner  |
| Account control       | Common device/IP accesses at least four linked accounts            | 25     | Fraud/device intelligence + AML |
| Spending anomaly      | At least three workers show $\geq 80\%$ transfer/cash depletion    | 14     | Transaction monitoring          |
| Geographic risk       | Three or more worker countries plus foreign cash-out               | 12     | Country-risk governance         |
| Sector risk           | Labour-intensive sector plus incomplete beneficial ownership       | 8      | KYC/EDD policy owner            |
| External intelligence | At least one high-risk linked-account intelligence hit             | 4      | Screening governance            |
| Document evidence     | At least three risky documents                                     | 4      | Document-control owner          |

#### Calibration rule

Weights and thresholds are prototype parameters. Production changes require documented rationale, labelled data, outcome analysis, versioning, regression testing, model-risk review, and approval.

## Decision transaction design

1. Validate the decision against the four allowed values.
2. Begin a database transaction.
3. Insert the named decision and rationale.
4. Insert an audit event referencing the generated decision ID.
5. Update the case status and assignee.
6. Commit all writes, or rollback all writes on error.

## Application query contract

All case-specific queries use bound parameters. The app attempts the active Snowflake session when hosted and falls back to `st.connection("snowflake").session()` locally. Cached reads have a short TTL; reviewer writes

bypass read caching and clear cached data after success.

Architecture decision record

| ADR    | DECISION                                     | REASON                                                               | CONSEQUENCE                                                            |
|--------|----------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------|
| ADR-01 | Snowflake is the system of analytical record | Keep evidence, logic, app reads, and audit lineage governed together | Local UI still depends on Snowflake connectivity                       |
| ADR-02 | Risk score is deterministic SQL              | Reproducibility and control take priority over opaque scoring        | Threshold changes require governed SQL releases                        |
| ADR-03 | Generative brief is optional                 | Core service must function without model access                      | Deterministic narrative may be less fluent but remains auditable       |
| ADR-04 | Reviewer decision and audit event are atomic | Prevent incomplete regulatory chronology                             | Any write failure rolls back the full decision                         |
| ADR-05 | Prototype uses synthetic evidence only       | Enable safe, reproducible demonstration                              | Production security and data controls remain separate acceptance gates |

16 • SOP

Standard Operating Procedure: investigate consistently, escalate safely.

SOP-01 • Start-of-day service check

- 1. Confirm the warehouse, app, source freshness, and latest successful load.
- 2. Confirm no failed data-quality or SQL test gates.
- 3. Review open cases, priorities, ownership, and SLA exceptions.
- 4. Escalate access, freshness, or integrity failures before case review.

SOP-02 • Investigator case review

| STEP                | INVESTIGATOR ACTION                                                       | EVIDENCE/CONTROL               | EXIT CONDITION                              |
|---------------------|---------------------------------------------------------------------------|--------------------------------|---------------------------------------------|
| 1. Orient           | Select the case and confirm customer/account context                      | Overview and case header       | Correct case and ownership confirmed        |
| 2. Validate signals | Review each detected typology and underlying values                       | Typology Signals               | No obvious data-quality explanation remains |
| 3. Trace network    | Follow employer, workers, controller, funnel, shared access, and cash-out | Account Network                | Material relationships understood           |
| 4. Corroborate      | Review transactions, documents, KYC, and external intelligence            | Evidence tabs and source links | Evidence confidence and gaps recorded       |
| 5. Challenge score  | Reconcile every point and route; note contrary evidence                   | Risk Score                     | Score accepted or challenged                |
| 6. Review brief     | Validate summary, hypothesis, missing evidence, and recommended action    | Case Intelligence Brief        | Narrative grounded in evidence              |
| 7. Decide           | Select allowed disposition, enter rationale, acknowledge human control    | Reviewer Decision              | Decision saved successfully                 |
| 8. Verify           | Confirm decision and linked audit event                                   | Audit Trail                    | Chronology complete                         |

### SOP-03 • Decision standards

| DECISION        | USE WHEN                                                | MINIMUM RATIONALE                                                         | NEXT ACTION                                   |
|-----------------|---------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------|
| Escalate to SAR | Multi-source suspicion meets internal escalation policy | Key facts, typologies, network, gaps, and why escalation is proportionate | Transfer to authorised SAR assessment process |
| Request more    | A material gap prevents a defensible disposition        | Specific missing item, owner, due date, and decision impact               | Return to evidence collection                 |

| DECISION                | USE WHEN                                                     | MINIMUM RATIONALE                                       | NEXT ACTION              |
|-------------------------|--------------------------------------------------------------|---------------------------------------------------------|--------------------------|
| evidence                |                                                              |                                                         |                          |
| Monitor case            | Concern exists but immediate escalation threshold is not met | Monitoring indicators, period, trigger, and owner       | Schedule enhanced review |
| Close as false positive | A documented benign explanation resolves the indicators      | Evidence-based explanation and residual-risk assessment | Close with audit record  |

### SOP-04 · Safeguarding and outreach

Potentially vulnerable customers

Do not contact a potentially controlled individual through a suspected controller's shared phone, email, device, or address. Follow the institution's safeguarding, confidentiality, SAR non-disclosure, and law-enforcement liaison procedures.

### SOP-05 · Exception and incident handling

1. Stop decision activity if evidence is stale, incomplete, mismatched, or exposed to an unauthorised person.
2. Record incident time, affected case/object, user, symptom, and immediate containment.
3. Route data defects to the data owner, app failures to engineering, and control/privacy issues to the relevant duty owner.
4. Resume only after documented validation; never reconstruct missing audit history manually without approval.

### 17 · CONTROL FRAMEWORK

Controls make the copilot defensible, not merely impressive.

| CONTROL ID | RISK                         | PREVENTIVE / DETECTIVE CONTROL             | EVIDENCE                       | FREQUENCY  |
|------------|------------------------------|--------------------------------------------|--------------------------------|------------|
| CTL-01     | Autonomous regulatory action | Human acknowledgement; no filing interface | Decision record and app design | Every case |

| CONTROL ID | RISK                     | PREVENTIVE / DETECTIVE CONTROL                                   | EVIDENCE                                       | FREQUENCY             |
|------------|--------------------------|------------------------------------------------------------------|------------------------------------------------|-----------------------|
| CTL-02     | Unexplained score        | Visible points, factors, version, and deterministic SQL          | Risk view and test output                      | Every calculation     |
| CTL-03     | Partial decision write   | Decision, audit, and case update in one transaction              | Stored procedure and linked IDs                | Every decision        |
| CTL-04     | Unsupported narrative    | Bounded evidence prompt and deterministic fallback               | Prompt text, source query, reviewer validation | Every brief           |
| CTL-05     | Unauthorised data access | Target RBAC, masking, secure views, access review                | Grant reports and access logs                  | Continuous / periodic |
| CTL-06     | Bad or stale data        | Contracts, freshness, referential, volume, and null tests        | Data-quality results                           | Every load            |
| CTL-07     | Threshold/model drift    | Versioned scoring, distribution/outcome monitoring, approval     | Calibration pack and change record             | Monthly / change      |
| CTL-08     | Sensitive-data leakage   | Synthetic prototype; production minimisation, masking, retention | Classification and retention evidence          | Design + continuous   |

### Target role model

| ROLE               | READ                | WRITE | RESTRICTED ACTION        |
|--------------------|---------------------|-------|--------------------------|
| SHADOWTRACE_VIEWER | Approved case views | None  | No raw restricted fields |

| ROLE                     | READ                             | WRITE                                              | RESTRICTED ACTION                       |
|--------------------------|----------------------------------|----------------------------------------------------|-----------------------------------------|
| SHADOWTRACE_INVESTIGATOR | Assigned case evidence           | Working notes / evidence requests in target design | Cannot approve escalation               |
| SHADOWTRACE_REVIEWER     | Full authorised case             | Allowed case decisions                             | Cannot alter score SQL or audit history |
| SHADOWTRACE_ENGINEER     | Technical metadata / masked data | Approved non-production objects                    | No business decision                    |
| SHADOWTRACE_ADMIN        | Platform administration          | Controlled deployment/configuration                | No silent production rule change        |

18 · OPERATING MODEL

Ownership spans business, risk, data, engineering, and assurance.

| ACTIVITY                     | PRODUCT OWNER | AML TYPOLOGY OWNER | ENGINEERING | REVIEWER / MLRO DELEGATE | MODEL RISK / AUDIT |
|------------------------------|---------------|--------------------|-------------|--------------------------|--------------------|
| Prioritise roadmap           | A/R           | C                  | C           | C                        | I                  |
| Define typologies/thresholds | C             | A/R                | C           | C                        | C                  |
| Build and deploy             | A             | C                  | R           | I                        | C                  |
| Review cases                 | I             | C                  | I           | A/R                      | I                  |
| Monitor outcomes/drift       | A             | R                  | R           | C                        | C                  |

| ACTIVITY                      | PRODUCT OWNER | AML TYPOLOGY OWNER | ENGINEERING | REVIEWER / MLRO DELEGATE | MODEL RISK / AUDIT |
|-------------------------------|---------------|--------------------|-------------|--------------------------|--------------------|
| Approve material model change | C             | R                  | C           | C                        | A                  |
| Assure controls               | C             | C                  | C           | C                        | A/R                |

R = Responsible · A = Accountable · C = Consulted · I = Informed. Named organisational owners must replace generic roles before production approval.

### Service cadence

#### Daily

Freshness, failures, backlog, SLA exceptions, high-risk cases, decision-write errors.

#### Weekly

Case outcomes, false positives, evidence gaps, investigator feedback, defect trends.

#### Monthly / change

Score distribution, drift, calibration, access review, control evidence, benefit realisation.

### RAID register and open decisions

| TYPE | ITEM                                                           | RESPONSE / DECISION NEEDED                                           | OWNER                 |
|------|----------------------------------------------------------------|----------------------------------------------------------------------|-----------------------|
| Risk | Country/sector features create unfair or stigmatising outcomes | Use only in combination; perform fairness and adverse-outcome review | Typology + model risk |

| TYPE       | ITEM                                                | RESPONSE / DECISION NEEDED                                                     | OWNER                    |
|------------|-----------------------------------------------------|--------------------------------------------------------------------------------|--------------------------|
| Risk       | Shared device/IP has benign explanations            | Require network and transaction corroboration; expose contrary evidence        | AML product              |
| Risk       | Generated narrative overstates suspicion            | Grounded prompt, deterministic fallback, reviewer validation, evaluation suite | AI governance            |
| Assumption | Stable account identifiers link the source domains  | Formalise data contracts and identity resolution                               | Data owner               |
| Dependency | External-intelligence licensing and permissible use | Confirm sources, retention, citations, and redistribution rights               | Compliance / procurement |
| Decision   | Jurisdiction and SAR hand-off integration           | Select target policy, authorised users, and filing boundary before production  | MLRO delegate            |
| Decision   | Production SLO, RTO, RPO, and retention periods     | Set from case criticality and enterprise policy                                | Service owner            |

19 • BENEFITS

# Build the business case from measured handling time and quality—not unsupported promises.

The calculator below is illustrative. Replace every default with baseline and pilot measurements from the target operating unit. Results estimate capacity value, not headcount removal or guaranteed savings.

Cases reviewed per month

500

Current minutes per case

120

Assisted minutes per case

80

Loaded analyst cost / hour

65

1,000

hours of monthly capacity released

\$7,80,000

illustrative annual capacity value

66.7%

active handling-time reduction

## Pilot KPI framework

| KPI                     | DEFINITION                                            | BASELINE SOURCE                 | PILOT SUCCESS METHOD         |
|-------------------------|-------------------------------------------------------|---------------------------------|------------------------------|
| Active handling time    | Analyst minutes actively spent per completed case     | Time study / workflow telemetry | Compare matched case cohorts |
| Time to first insight   | Alert creation to first evidence-backed hypothesis    | Case timestamps                 | Median and 90th percentile   |
| Evidence completeness   | Required sources assessed before decision             | Quality sample                  | Percentage meeting checklist |
| Decision rework         | Cases returned or reopened for missing/weak evidence  | Case outcomes                   | Rate and reason distribution |
| Explainability coverage | Score points with accessible evidence and rationale   | Automated trace test            | Target defined by governance |
| Audit completeness      | Decisions with actor, time, rationale, object linkage | Audit query                     | Percentage complete          |
| Investigator acceptance | Usefulness and trust rating from structured UAT       | Survey / interviews             | Agreed acceptance threshold  |

### Benefits realisation method

Establish the current-state baseline, run a controlled pilot with comparable case types, measure quality and time together, review adverse outcomes, then approve any scaled benefit claim.

# Testing covers data, logic, workflow, security, usability, and resilience.

| TEST LAYER           | MINIMUM COVERAGE                                                    | PROTOTYPE EVIDENCE            | PRODUCTION GATE                  |
|----------------------|---------------------------------------------------------------------|-------------------------------|----------------------------------|
| Static / packaging   | Python compile, HTML integrity, expected files                      | Local validation              | CI pipeline                      |
| Data quality         | Schema, required fields, JSON, referential links, duplicate keys    | CSV validation                | Automated load quarantine        |
| Rule unit tests      | Positive, negative, boundary for each typology                      | C003 positive + C010 negative | Versioned test fixtures          |
| Score regression     | Score, band, evidence strength, route, breakdown                    | Six SQL assertions            | Golden-case suite                |
| Workflow integration | All tabs, decision options, rollback, cache refresh                 | Demo smoke test               | Automated end-to-end suite       |
| Security             | Role isolation, masking, injection, secrets, access logs            | Design only                   | Mandatory security test          |
| Performance          | Interactive case load and concurrent reviewer volume                | Not benchmarked               | Load test against SLO            |
| UAT                  | Realistic workflow, narrative quality, explainability, safeguarding | Demo script                   | Named investigator/risk sign-off |

## Release acceptance criteria

- ☐ All ordered SQL files execute without error in the target environment
- ☐ All automated and golden-case tests pass
- ☐ CASE-C003 reconciles to 92 and CASE-C010 remains LOW
- ☐ Every allowed decision writes exactly one linked audit event

- ☐ Human gate, access, data, prompt, and rollback controls are evidenced
- ☐ Deployment, rollback, incident, and service-owner details are current

## 21 • HACKATHON EVIDENCE

# Use CoCo CLI deliberately

CoCo's executable is `cortex`. Run it in plan mode from the repository root so Snowflake writes remain reviewable.

```
cortex -c shadowtrace -w . --plan
```

## Recommended activities

1. Inspect live schema drift.
2. Execute each typology for CASE-C003.
3. Reconcile all 92 score points.
4. Review the Streamlit implementation.
5. Debug failed Snowflake queries.
6. Summarize evidence with query IDs.

## Evidence to retain

- Prompt and response screenshot
- Snowflake query ID
- Before/after diff for a real correction
- Six PASS test output
- No secrets or personal data visible

### Prompt source

Use the prepared prompt journal in `coco/prompts.md` and workflow in `coco/workflow.md`. Do not fabricate output; capture the connected session.

## 22 • PRESENTATION

# Run the five-minute happy-path demo

| TIME | ACTIVITY                                       | JUDGE-VISIBLE PROOF             |
|------|------------------------------------------------|---------------------------------|
| 0:00 | State the fragmented AML investigation problem | One-line value proposition      |
| 0:35 | Run the typology query in Snowsight            | Five detected rows              |
| 1:10 | Open Streamlit and select CASE-C003            | 92/100 CRITICAL                 |
| 1:45 | Trace employer → workers → controller → funnel | Interactive network graph       |
| 2:30 | Show documents and external intelligence       | Multi-source evidence           |
| 3:10 | Explain every score component                  | Transparent 92-point breakdown  |
| 3:45 | Show the case brief and grounded prompt        | Actionable copilot narrative    |
| 4:20 | Record Escalate to SAR                         | Named human decision gate       |
| 4:45 | Open Audit Trail                               | Linked decision and audit event |

Reviewer input

Reviewer: AML Demo Reviewer

Decision: Escalate to SAR

Rationale: Multi-source evidence supports escalation for formal SAR assessment.

Acknowledgement: Checked

Use careful language

Say "indicators," "suspicion hypothesis," and "requires investigation." Do not state that exploitation or guilt has been proven.

23 · SNOWFLAKE-NATIVE DEPLOYMENT

# Deploy Streamlit inside Snowflake

Local execution is sufficient for development. For the strongest submission, deploy the app natively.

## 1 Set the correct warehouse

Edit `sql/07_deploy_streamlit.sql` and replace `COMPUTE_WH` if your warehouse has another name.

## 2 Deploy

```
snow sql -c shadowtrace -f sql/07_deploy_streamlit.sql
```

## 3 Open in Snowsight

Navigate to Streamlit and open `SHADOWTRACE_AI.AML.SHADOWTRACEAI_APP`. Re-run the happy path in the hosted application.

## Production deployment runbook

| PHASE      | REQUIRED ACTIVITY                                                                          | EVIDENCE / OWNER      |
|------------|--------------------------------------------------------------------------------------------|-----------------------|
| Pre-deploy | Approved change, reviewed diff, passing tests, backup/rollback point, warehouse/cost check | Release owner         |
| Deploy     | Apply versioned schema changes before dependent views/app; capture query IDs               | Engineering           |
| Smoke      | Load C003/C010, open all tabs, record non-regulatory demo decision, verify audit linkage   | Engineering + product |
| Observe    | Monitor errors, query latency, freshness, score distribution, decision writes              | Service owner         |
| Release    | Communicate version, known limitations, support contact, and evidence pack                 | Product owner         |

## Rollback principle

Stop new reviewer decisions, restore the last approved app/view version, validate core reads and audit writes, reconcile any in-flight case activity, document impact, and reopen only after owner approval. Never delete production audit history as a rollback mechanism.

### 24 • SUBMISSION PACKAGE

## Capture credible execution evidence

Screenshots should prove SQL, data, application, reviewer action, and audit persistence—not merely show a visual mockup.

- ☐ Snowflake tables and views under SHADOWTRACE\_AI . AML
- ☐ Five CASE-C003 typology result rows
- ☐ Streamlit Overview showing 92/100 CRITICAL
- ☐ Employer/worker/controller/funnel network
- ☐ Documents and external intelligence
- ☐ Explainable risk breakdown
- ☐ Case intelligence brief
- ☐ Reviewer decision and resulting audit row
- ☐ CoCo CLI output with query ID

#### Recommended format

Capture at 1440×900 or larger, keep the case ID visible, hide credentials, and use the filenames specified in docs/screenshots/README.md.

### 25 • RECOVERY GUIDE

## Troubleshooting

► snow is not recognized

► PowerShell blocks virtual-environment activation

► Snowflake authentication or MFA repeatedly fails

► **CREATE DATABASE or CREATE STREAMLIT is denied**

► **The CSV PUT command cannot find files**

► **Streamlit shows “connection or setup is incomplete”**

► **CASE-C003 is not 92**

► **The optional Cortex view fails**

► **Plotly or Streamlit dependency error**

## 26 · REPEATABLE DEMOS

# Reset the reviewer activity

Use this only on the synthetic demo database.

```
DELETE FROM SHADOWTRACE_AI.AML.audit_events
WHERE case_id = 'CASE-C003' AND event_type = 'REVIEWER_DECISION';

DELETE FROM SHADOWTRACE_AI.AML.case_decisions
WHERE case_id = 'CASE-C003';

UPDATE SHADOWTRACE_AI.AML.case_alerts
SET alert_status = 'OPEN', assigned_to = 'AML_DEMO_REVIEWER'
WHERE case_id = 'CASE-C003';
```

### Scope warning

Do not run reset SQL against real AML data. The commands intentionally target the synthetic CASE-C003 identifier.

## 27 · COMPLETION GATE

# Final activity checklist

## Technical

- ☐ Schema and seed load completed
- ☐ All six SQL tests pass
- ☐ Local app works across eight tabs
- ☐ Decision creates linked audit event
- ☐ Snowflake-native app deployed

## Submission

- ☐ Real CoCo activity captured
- ☐ Screenshot set complete
- ☐ Five-minute demo rehearsed
- ☐ Human gate and responsible-use language included
- ☐ No credentials committed or visible

### Finish line

You are ready when the judges can see the source data, execute the SQL, understand every point in 92, navigate the evidence network, record a human decision, and verify its audit event.

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**ShadowTraceAI: AML Intelligence Copilot on Snowflake**

Synthetic demonstration project · Domain-Specific AI Copilot track · Generated 25 July 2026

Repository: C:\Users\manoj\Development\Github\ShadowTraceAI-Snowflake-CoCo